

1 Genome sequencing

1.1 Extraction of genome DNA

Genomic DNA was extracted with the SDS method^[1]. The harvested DNA was detected by the agarose gel electrophoresis and quantified by Qubit® 2.0 Fluorometer (Thermo Scientific).

1.2 Library construction

(1) PacBio Sequel platform

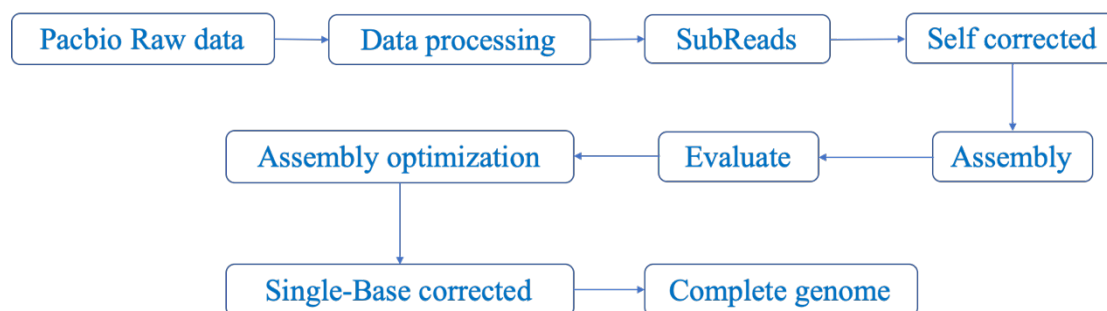
Libraries for single-molecule real-time (SMRT) sequencing was constructed with an insert size of 10 kb using the SMRT bell TM Template kit, version 1.0. Briefly, the process were that fragment and concentrate DNA, repair DNA damage and ends, prepare blunt ligation reaction, purify SMRTbell Templates with 0.45X AMPure PB Beads, size-selection using the BluePippin System, repair DNA damage after size-selection. At last, the library quality was assessed on the Qubit® 2.0 Fluorometer (Thermo Scientific) and detected the insert fragment size by Agilent 2100(Agilent Technologies).

(2) Illumina NovaSeq platform

A total amount of 1µg DNA per sample was used as input material for the DNA sample preparations. Sequencing libraries were generated using NEBNext® Ultra™ DNA Library Prep Kit for Illumina (NEB, USA) following manufacturer's recommendations and index codes were added to attribute sequences to each sample. Briefly, the DNA sample was fragmented by sonication to a size of 350bp, then DNA fragments were end-polished, A-tailed, and ligated with the full-length adaptor for Illumina sequencing with further PCR amplification. At last, PCR products were purified (AMPure XP system) and libraries were analysed for size distribution by Agilent2100 Bioanalyzer and quantified using real-time PCR.

1.3 Sequencing

The whole genome of XXX was sequenced using PacBio Sequel platform and Illumina NovaSeq PE150 at the Beijing Novogene Bioinformatics Technology Co., Ltd.



2 Genome assembly

2.1 Preliminary assembly with SMRT Link v5.0.1

1) In order to ensure the accuracy of the subsequent analysis results, the low-quality reads were filtered (less than 500 bp) to obtain Clean data;

2) Using the automatic error correction function of SMRT portal, the long reads were selected (more than 6000bp) as the seed sequence, and the other shorter reads were aligned to the seed sequence by Blasr, so that the accuracy of the seed sequence could be improved further. After assembling we got an initial result.

2.2 Correct the results of the preliminary assembly

By the variant Caller module of the SMRT Link software, the arrow algorithm was used to correct and count the variant sites in the preliminary assembly results.

2.3 Correct the assembly result with Illumina data

The corrected assembly result, which was used as the reference sequence, was blast with Illumina data by bwa. Furthermore the result was filtered with the base minimum mass value of was 20 , the minimum read depth of 4 and the maximum read depth of 1000.

2.4 Cyclization, initiation site correction

Based on the overlap between the head and the tail, we confirmed whether the chromosomal sequence formed a circle or not, then corrected the initial site by blast with the DNAA database. At last the chromosome and plasmid sequences were screened by blast with plasmid database.

3 Genome Component prediction

Genome component prediction included the prediction of the coding gene, repetitive sequences,

non-coding RNA, genomics islands, transposon, prophage, and clustered regularly interspaced short palindromic repeat sequences (CRISPR). The available steps were proceeded as follows:

- 1) For bacteria, we used the GeneMarkSprogram to retrieve the related coding gene.
- 2) The interspersed repetitive sequences were predicted using the RepeatMasker (<http://www.repeatmasker.org/>). The tandem Repeats were analyzed by the TRF (Tandem repeats finder).
- 3) Transfer RNA (tRNA) genes were predicted by the tRNAscan-SE . Ribosome RNA (rRNA) genes were analyzed by the rRNAmmer .Small nuclear RNAs(snRNA)were predicted by BLAST against the Rfam database.
- 4) The IslandPath-DIOMB program was used to predict the Genomics Islands, and transposon PSI was used to predict the transposons based on the homologous blast method. The PHAST was used for the prophage prediction (<http://phast.wishartlab.com/>) and the CRISPRFinder was used for the CRISPR identification.

4 Gene function

We used seven databases to predict gene functions. They were respective GO (Gene Ontology), KEGG (Kyoto Encyclopedia of Genes and Genomes), COG (Clusters of Orthologous Groups), NR (Non-Redundant Protein Database databases), TCDB (Transporter Classification Database), and, Swiss-Prot . A whole genome Blast search (E-value less than $1e-5$, minimal alignment length percentage larger than 40%) was performed against above seven databases. The secretory proteins were predicted by the Signal P database, and the prediction of Type I-VII proteins secreted by the pathogenic bacteria were based on the EffectiveT3 software. Meanwhile, we analyzed the secondary metabolism gene clusters by the antiSMASH. For pathogenic bacteria, we added the pathogenicity and drug resistance analyses. We used the PHI (Pathogen Host Interactions), VFDB (Virulence Factors of Pathogenic Bacteria), ARDB (Antibiotic Resistance Genes Database) to perform the above analyses. Carbohydrate-Active enzymes were predicted by the Carbohydrate-Active enZymes Database.

5 Comparative genomics analysis

Comparative genomic analysis included the genomic synteny, the core genes and specific genes, gene family phylogenetic tree, SNP (Single Nucleotide Polymorphism), indel (insertion and deletion) and SV (Structural Variation) annotation, and genome visualization. 1) Genomic alignment between the sample genome and reference genome (or among more than two sample genomes) were performed using the MUMmer and LASTZ tools. Genomic synteny was analyzed based on the alignment results. 2) Core genes and specific genes were analyzed by the CD-HIT rapid clustering of similar proteins software with a threshold of 50% pairwise identity and 0.7 length difference cutoff in amino acid. Then the venn figure was drawn to show their relationships among the samples. 3) Gene family was constructed by the gene of XXX and XXX, using the multiple softwares: Blast was used to pairwise align all genes and eliminate the redundancy by solar and carried out gene family clustering treatment based on the alignment results with Hcluster_sg software. 4) The phylogenetic tree was constructed by the TreeBeST or PhyML and the setting of bootstraps was 1,000 with the orthologous genes. 5) SNP, indel and SV were found by the genomic alignment results among samples by the MUMmer and LASTZ we mentioned them before. 6) Genome overview was created by Circos to show the annotation information.

Reference

[1] Lim H J, Lee, E. - H, Yoon Y, et al. Portable lysis apparatus for rapid single - step DNA extraction of *Bacillus subtilis*[J]. Journal of Applied Microbiology, 2016, 120(2):379-387.